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Accounting Identities Are Not Evidence: A Formal Proof That Definitional Truths Cannot Support Causal Claims

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Abstract

An accounting identity is a statement whose truth follows from definitions and conservation laws alone. Familiar examples include the national-income identity, the double-entry bookkeeping identity, the caloric energy-balance equation, and the atmospheric carbon mass balance $F_{\text{anth}} - \mathrm{d}M_a/\mathrm{d}t = F_{\text{out}} - F_{\text{nat}}$. We prove, using only elementary set theory, that no accounting identity can serve as evidence for any claim about cause. Two consequences follow at once. First, the identity cannot pick a side between a causal claim and its reverse. Second, stacking many accounting identities does not help: the overlap of any finite collection of identities is again the whole state space, so the "many lines of evidence" defense is empty when every line is an identity rewritten in different notation. The same error has been named independently across economics, medicine, demography, fluid dynamics, and double-entry bookkeeping. The carbon mass balance is one instance. The Global Carbon Budget, IPCC AR6 Working Group I Chapter 5, and the Summary for Policymakers' "about 56%" land-and-ocean uptake figure each present a rearrangement of the mass-balance identity as evidence for human causation. The non-peer-reviewed EHB (2024) report states openly that the mass balance "definitely proves" human causation. Five separate presentations of the identity in that paper are audited and shown to be empty. The paper's stated conclusion is invalidated by the theorem applied to the very argument advanced for it.

Keywords: accounting identity; causal inference; carbon cycle; mass balance; atmospheric CO_2 attribution; airborne fraction; Bayesian confirmation; definitional truth; anthropogenic attribution

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1. Introduction

Accounting identities are statements whose truth follows from definitions and conservation laws alone. Two familiar examples are the double-entry bookkeeping identity Assets ≡ Liabilities + Equity and the national-income identity $Y \equiv C + I + G + NX$, where:

- Y = Gross Domestic Product (GDP): the total market value of final goods and services produced domestically during a given period.
- C = Consumption: spending by households on goods and services.
- I = Investment: spending on capital equipment, structures, residential construction, and changes in business inventories.

- G = Government purchases: government spending on goods and services, excluding transfer payments such as Social Security.
- NX = Net exports: exports minus imports ($X - M$).

This paper proves one general fact about accounting identities. No accounting identity can serve as evidence for any claim about cause. The proof uses only elementary set theory.

The result is general. Section 7 shows the same error named independently in economics, medicine, demography, fluid dynamics, and double-entry bookkeeping. The carbon mass balance is one instance. We treat it in detail in Sections 8 and 10.

The carbon-cycle instance arises as follows. The fundamental starting point is conservation of mass in the atmospheric reservoir:

$$\frac{\mathrm{d}M_a}{\mathrm{d}t} \;=\; F_{\mathrm{in}} - F_{\mathrm{out}}, \tag{1}$$

where M_a is the atmospheric carbon mass and $F_{\mathrm{in}}, F_{\mathrm{out}}$ are the total gross input and output fluxes. Equation (1) says one thing. The mass of CO_2 in the atmosphere changes at a rate equal to total gross inflow minus total gross outflow. Only the left side is a measured quantity. M_a is observed directly by atmospheric sampling at Mauna Loa, the NOAA Global Monitoring Division network, and equivalent stations. The rate $\mathrm{d}M_a/\mathrm{d}t$ follows by numerical differentiation of that record. Neither F_{in} nor F_{out} is directly measured anywhere on Earth. They are definitional quantities. Their values are not fixed by observation. They are fixed by the bookkeeping scheme itself.

The bookkeeping scheme then splits the input flux into a non-fossil-fuel part and a fossil-fuel part:

$$F_{\mathrm{in}} \;\equiv\; F_{\mathrm{nat}} + F_{\mathrm{anth}}. \tag{2}$$

This split is a bookkeeping choice, not a measurement. F_{anth} is estimated from bottom-up inventories of fossil-fuel combustion, cement production, and land-use change. F_{nat} is not measured anywhere. It enters only through the split, as whatever is left over.

By "bookkeeping scheme" we mean this: the conservation equation (1) plus the choice to split the input flux into a human part and a non-human part per equation (2). That pair of statements is the whole scheme. It is a way of keeping accounts. It is not, by itself, a claim about what causes what.

Substituting (2) into (1) and rearranging yields the mass-balance relation in its commonly cited form:

$$F_{\mathrm{anth}}(t) - \frac{\mathrm{d}M_a}{\mathrm{d}t} \;=\; F_{\mathrm{out}}(t) - F_{\mathrm{nat}}(t). \tag{3}$$

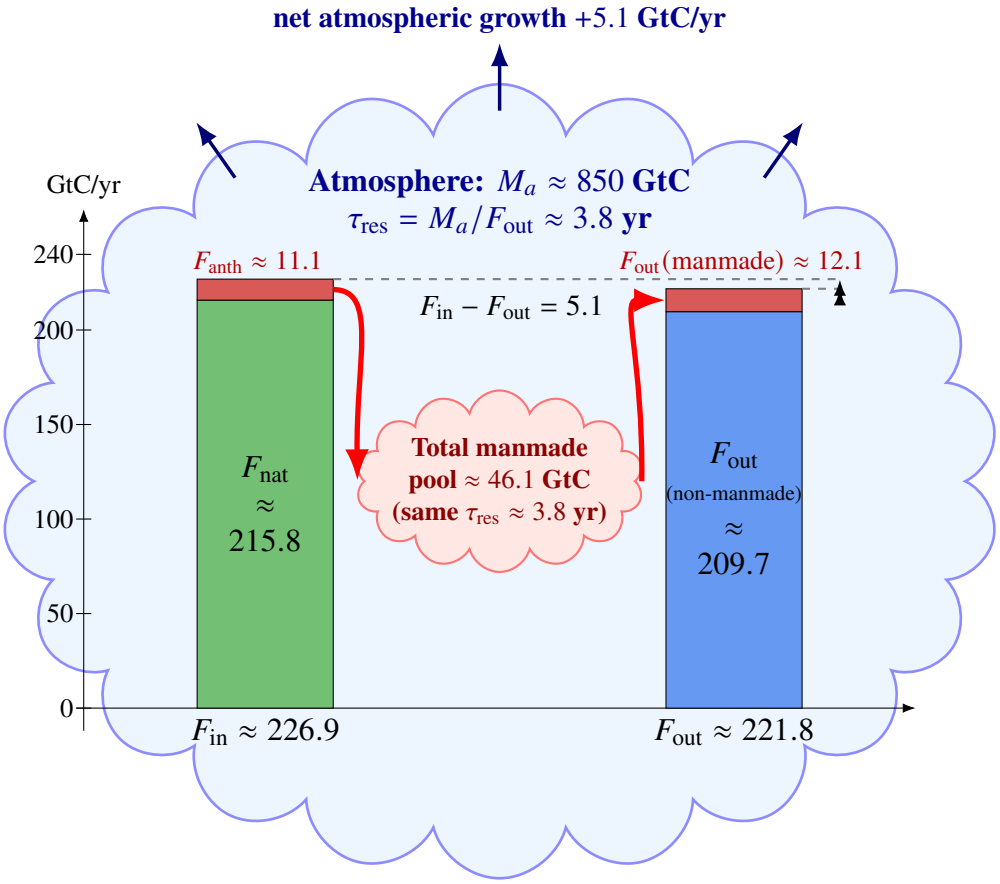
Equation (3) adds no empirical content to (1) and (2). It is an algebraic rearrangement of the postulate and the definition. It holds in every situation in which the postulate and the definition hold. Nothing has been measured in going from (1) and (2) to (3). The two unknowns on the right side are not pinned down by this rearrangement. Only their difference is. Any situation in which humans emit F_{anth}, nature exchanges F_{nat} and F_{out}, and the atmosphere grows at $\mathrm{d}M_a/\mathrm{d}t$ consistent with (1) satisfies (3) automatically.

The claim proved below is that no equation of this form can serve as evidence, in any sense, for any claim about cause. The result is elementary. Its persistent violation in the applied literature warrants a formal statement.

The peer-reviewed literature that anchors the anthropogenic-attribution consensus commits this error openly. Ballantyne et al. (2012) [1] in *Nature*: "a comprehensive carbon dioxide mass balance analysis shows that net global carbon uptake has increased by about 0.05 billion tonnes per year over the past 50 years and that in that time the global carbon uptake has almost doubled." The "carbon uptake" quantity is $F_{\mathrm{anth}} - \mathrm{d}M_a/\mathrm{d}t$: identity (3) rearranged, presented as "measurement." The annual Global Carbon Budget of Friedlingstein et al. (2026) [10] makes the identity the

equation that closes its budget, and reports its residual as "the carbon budget imbalance (BIM), a measure of imperfect data and incomplete understanding of the contemporary carbon cycle." The identity is not evidence for attribution in that scheme; it is the scheme. Canadell et al. (2021) [2], in the Executive Summary of Chapter 5 of the IPCC AR6 Working Group I report, asserts that "it is unequivocal that the increases in atmospheric carbon dioxide (CO_2), methane (CH_4) and nitrous oxide (N_2O) since the pre-industrial period are overwhelmingly caused by human activities," and the Summary for Policymakers of the same report supplies the number: "land and ocean have taken up a near-constant proportion (globally about 56% per year) of CO_2 emissions from human activities over the past six decades." The "56%" figure is the sink-uptake ratio $(F_{\text{anth}} - \mathrm{d}M_a/\mathrm{d}t)/F_{\text{anth}}$, equivalently $1 - (\mathrm{d}M_a/\mathrm{d}t)/F_{\text{anth}}$: identity (3) rearranged, presented as "attribution." The most explicit invocation appears in the non-peer-reviewed EHB (2024) report of Engelbeen et al. (2024) [8]. The opening of its central chapter states plainly: "The carbon mass balance is what definitely proves that humans are the main cause of the increase of CO_2 in the atmosphere: as long as the increase in the atmosphere is less than what humans emit per year, nature as a whole is a net sink and can't be the cause of the overall increase because that would violate the mass balance." The claim reduces attribution of the atmospheric CO_2 increase to an appeal to the identity itself, which is precisely the inference Theorem 1 forbids.

Figure 1 previews the geometric point by decomposing the atmospheric CO_2 flux budget by origin (manmade versus non-manmade), using 2010–2019 decadal averages from Canadell et al. (2021) [2] and Friedlingstein et al. (2026) [10]. The gross input to the atmosphere is $F_{\text{in}} = F_{\text{anth}} + F_{\text{nat}} \approx 226.9$ GtC yr^{-1} and the gross output is $F_{\text{out}} \approx 221.8$ GtC yr^{-1}, so the net atmospheric gain is $\mathrm{d}M_a/\mathrm{d}t = F_{\text{in}} - F_{\text{out}} \approx 5.1$ GtC yr^{-1}. The manmade-origin portion of each bar is shown in red; the non-manmade portion in green (input side) or blue (output side). Total manmade emission $F_{\text{anth}} \approx 11.1$ GtC yr^{-1} (fossil-fuel combustion plus land-use change minus the cement-carbonation sink, from the Global Carbon Budget) appears as the red top of F_{in}. The whole atmospheric reservoir ($M_a \approx 850$ GtC) is drawn as a cloud wrapping the bars; a smaller sub-cloud inside represents the manmade sub-pool (≈ 46.1 GtC), computed as the 2010–2019 decadal average of the convolutional integral of the GCB total anthropogenic series back to 1959 under residence-time decay $\tau_{\text{res}} = M_a/F_{\text{out}} \approx 3.8$ yr. Dividing the pool by the same τ_{res} gives the manmade outflow ≈ 12.1 GtC yr^{-1} (red top of F_{out}). Red pipes trace manmade carbon through the atmosphere: F_{anth} in, through the manmade sub-pool, back out. Outward arrows on the atmospheric cloud boundary indicate the net atmospheric growth, ≈ 5.1 GtC yr^{-1}. The non-manmade portions and the overall gross flux magnitudes are not fixed by observation; the identity (3) constrains only the difference $F_{\text{out}} - F_{\text{nat}}$, which must equal $F_{\text{anth}} - \mathrm{d}M_a/\mathrm{d}t \approx 6.0$ GtC yr^{-1}. The identity by itself cannot say which origin is driving atmospheric growth. The decomposition in the figure makes the empirical story sharp. Manmade outflow (≈ 12.1) *exceeds* manmade inflow ($F_{\text{anth}} \approx 11.1$), so manmade carbon was a *net sink* of about -1.0 GtC yr^{-1} over 2010–2019. Non-manmade carbon was a *net source* of $F_{\text{nat}} - F_{\text{out}}$(non-manmade) $\approx 215.8 - 209.7 \approx 6.1$ GtC yr^{-1}. Non-manmade sources therefore accounted for more than the entire 5.1 GtC yr^{-1} atmospheric gain, and manmade activity slightly offset it ($-1.0 + 6.1 = 5.1$). This is the opposite of the IPCC and EHB framing in which nature is presented as a net sink absorbing part of human emissions. The identity (3) is satisfied in both framings (box at the bottom of Figure 1), which is precisely why the identity cannot be used to choose between them.



Mass balance, Eq. (3): $F_{anth} - dM_a/dt = F_{out} - F_{nat}$

$11.1 - 5.1 = 221.8 - 215.8 = 6.0$ **GtC yr**$^{-1}$ ✓

By origin: Nature $+6.1$ **+ Manmade** $-1.0 = +5.1$ **GtC yr**$^{-1} = dM_a/dt$ ✓

Figure 1: Composition of the atmospheric CO_2 flux budget by origin (manmade vs. non-manmade), averaged over 2010–2019; all fluxes in GtC yr^{-1}. Decadal averages $M_a \approx 850$ GtC, $F_{in} \approx 226.9$, and $F_{out} \approx 221.8$ are from Canadell et al. (2021) [2] (IPCC AR6 Fig. 5.12); see also Cohler & Soon (2026b) [6] Table 11 for additional derivations and discussion. Total anthropogenic emission $F_{anth} \approx 11.1$ (fossil-fuel plus land-use changes minus cement carbonation) is the 2010–2019 Global Carbon Budget average of Friedlingstein et al. (2026) [10]. The manmade sub-pool (inner pink cloud) ≈ 46.1 GtC is the decadal-average convolutional integral of the GCB total anthropogenic series back to 1959 under residence-time decay $\tau_{res} = M_a/F_{out} = 850/221.8 \approx 3.8$ yr; the manmade outflow follows as $46.1/3.8 \approx 12.1$. Left bar (F_{in}): $F_{nat} \approx 215.8$ (green) + $F_{anth} \approx 11.1$ (red top). Right bar (F_{out}): non-manmade outflow ≈ 209.7 (blue) + manmade outflow ≈ 12.1 (red top). Red pipes trace manmade carbon through the atmosphere; the gap between bar tops and the outward cloud-boundary arrows both show $dM_a/dt = 5.1$. Key insight: manmade outflow (12.1) exceeds F_{anth} (11.1), so manmade is a net sink of -1.0; the entire growth plus that shortfall is non-anthropogenic, $215.8 - 209.7 = 6.1$. The boxed equation shows the mass-balance identity (3) is satisfied, as it must be in every allowed situation. The identity pins down only this one difference, not F_{nat} or F_{out} individually; its satisfaction therefore carries no causal information. That is Theorem 1.

Zero free parameters. Every quantity in Figure 1 is either a direct measurement, a published inventory, or a definition applied to a measurement. $M_a \approx 850$ GtC is the Mauna Loa concentration record times the standard conversion factor 2.124 GtC/ppm, as reported by Canadell et al. (2021) [2] Figure 5.12. F_{in} and F_{out} are from the same source. F_{anth} is the Global Carbon Budget anthropogenic series of Friedlingstein et al. (2026) [10]. dM_a/dt is computed as the 5-point Savitzky-Golay centered derivative of the Mauna Loa concentration for each year 2010–2019 [6], converted to

GtC and averaged; the result closes with the independent AR6 flux difference $F_{\text{in}} - F_{\text{out}}$ to within rounding. $F_{\text{nat}} = F_{\text{in}} - F_{\text{anth}}$ follows from the bookkeeping split (2). The residence time $\tau_{\text{res}} = M_a / F_{\text{out}}$ is the standard hydrological definition of outflow timescale, confirmed independently by Koutsoyiannis (2024a) [13] from the Mauna Loa and Barrow concentration relaxations. The manmade sub-pool ≈ 46.1 GtC is the convolutional integral of the GCB anthropogenic series back to 1959 under exponential decay with this same τ_{res}; nothing is fit. The manmade outflow ≈ 12.1 GtC yr^{-1} follows as pool divided by τ_{res}, because CO_2 molecules are chemically indistinguishable by origin and the atmosphere is well-mixed, so every sub-pool decays at the same rate as the bulk. Non-manmade outflow is then $221.8 - 12.1 = 209.7$ by subtraction. No external carbon-cycle construct enters the figure: no pulse-response function, no asserted long tail from ocean surface chemistry, and no fast-vs-slow sub-cycle partitioning. Any such construct would require manmade CO_2 molecules to decay at a different rate than non-manmade CO_2 molecules, which chemistry forbids in a well-mixed reservoir. The data, the math, and the physics leave no question: non-manmade sources drove the entire atmospheric CO_2 growth of the 2010s; manmade sources were a net sink for the period.

2. Notation and Basic Conventions

We use only elementary set-theory notation. The symbols are listed here for easy reference.

- Ω: the collection of all possible situations we consider (the "state space").
- $\omega \in S$: the situation ω belongs to the collection S.
- $S \subseteq \Omega$: S is a part of Ω (possibly all of it).
- $S \subsetneq \Omega$: S is a part of Ω, and Ω contains something S does not.
- $\varnothing$: the empty collection, containing nothing.
- $A \cap B$: the overlap of A and B, the collection of situations that belong to both.
- $A \cup B$: the union of A and B, the collection of situations that belong to one or the other or both.
- $\neg C$: the collection of situations in Ω that do not belong to C (the complement of C in Ω).

Three facts we use repeatedly, each a one-line consequence of the definitions above:

1. $\Omega \cap S = S$ for any S. Overlapping anything with the whole set changes nothing.
2. $S \cap S = S$.
3. $\neg(\neg C) = C$.

That is the entire math the main theorem needs. Everything else is English.

3. Definitions

We now define the terms the theorem uses. Each definition is first stated in plain English, then in formal notation.

Situation and state space

A *situation* is one possible state of the system under study. A situation lists the values of every quantity the bookkeeping scheme names. The *state space*, written Ω, is the collection of all possible situations.

For the carbon mass-balance application, a situation is the record over time of four quantities: all fossil-fuel CO_2 emissions $F_{\text{anth}}(t)$, all non-fossil-fuel CO_2 emissions $F_{\text{nat}}(t)$, all CO_2 absorbed

from the atmosphere by sinks $F_{\text{out}}(t)$, and the current mass of CO_2 in the atmosphere $M_a(t)$. We call that record ω, and write it as

$$\omega \;=\; \{F_{\text{anth}}(t),\, F_{\text{nat}}(t),\, F_{\text{out}}(t),\, M_a(t)\}_{t\in T}.$$

The subscript $t \in T$ reads "for every time t in the span of years T we are considering." A situation is allowed only if both conservation (1) and the fossil/non-fossil split (2) hold at every time t.

Within each situation we distinguish two kinds of quantity.

- The *measured* quantities are fixed from outside the bookkeeping scheme by direct observation or by an independent inventory. For the mass balance, these are M_a (measured atmospheric carbon mass) and F_{anth} (bottom-up inventory of fossil-fuel combustion, cement production, and land-use change).
- The *definitional* quantities are introduced by the bookkeeping scheme itself and are not fixed from outside, because there is no way to measure them empirically. For the mass balance, these are F_{nat} and F_{out}. This is why the Global Carbon Budget of Friedlingstein et al. (2026) [10] cannot report them as observations: it either omits them, invokes models and assumptions to supply values for them, or treats their difference as whatever closes the budget by arithmetic.

We write $\mathbf{m}(\omega)$ for the measured quantities in situation ω and $\mathbf{d}(\omega)$ for the definitional quantities. In plain words: the letter $\mathbf{m}$ stands for "measured" and stands in for the pair $(M_a(t), F_{\text{anth}}(t))$ taken over all times in the situation. The letter $\mathbf{d}$ stands for "definitional" and stands in for the pair $(F_{\text{nat}}(t), F_{\text{out}}(t))$ taken over all times in the situation.

Claim (proposition)

A *claim* is a yes-or-no statement about which situation we are in. Every claim picks out the collection of situations in which the claim is true. That collection is a part of Ω. We identify the claim with its collection.

Accounting identity

Definition 1 (Accounting identity)**.** *A claim E is an* accounting identity *when $E = \Omega$. Equivalently, the equation defining E holds in every situation, by definitions and conservation alone.*

In plain words: an accounting identity is a claim that is true no matter which situation we are in. It rules out no situation. That is what makes it an identity. That is also what will make it useless as evidence.

Observation

Definition 2 (Observation)**.** *A claim O is an* observation *when O fixes the measured quantities to their observed values and leaves the definitional quantities free. In symbols, $O = \{\omega \in \Omega : \mathbf{m}(\omega) = \mathbf{m}^*\}$ for some observed value $\mathbf{m}^*$.*

In plain words: O is the collection of all those situations in which the measured quantities take exactly the values we actually observed. The notation reads: "the set of all situations ω drawn from Ω such that the measured quantities in ω equal the observed value $\mathbf{m}^*$." The star on $\mathbf{m}^*$ marks it as the specific value that was measured in the real world, as opposed to the general variable $\mathbf{m}(\omega)$ that can take any value from one situation to the next. The definitional quantities F_{nat} and F_{out} are not constrained by the observation; they can still take any values consistent with ω being allowed.

An observation rules out every situation in which the measured quantities take values different from what was observed. This is precisely Popperian falsification in set-theoretic form [16]. In plain words: a claim earns scientific content by excluding possible situations. The more situations

a claim rules out, the more it says. An observation inconsistent with a theory falsifies that theory by excluding every situation the theory allowed. An observation is therefore a proper part of Ω (that is, $\varnothing \subsetneq O \subsetneq \Omega$), not all of it. An accounting identity, by contrast, rules out no situation at all, so it falsifies nothing and carries no information about which situation we are in. In Popper's terms it has zero empirical content. This is the first place at which observations part company with identities.

Contingent and causal claims

Definition 3 (Contingent claim)**.** *A claim C is* contingent *when $\varnothing \subsetneq C \subsetneq \Omega$. In plain words: C picks out some but not all situations. The empty collection $\varnothing$ sits strictly below C, meaning at least one situation does make C true. The full state space Ω sits strictly above C, meaning at least one situation makes C false.*

Definition 4 (Causal claim)**.** *A* causal claim *is a contingent claim that asserts a directed dependence between physical quantities (for example, "quantity X drives quantity Y"). The theorem below needs only that C be contingent; the intended application is to causal C.*

4. The Main Theorem

Before any symbols, the theorem in one sentence. If a claim E is true in every situation, then learning E rules out no situation. It therefore cannot turn any other claim C from true to false or from false to true, and it cannot narrow the set of situations in which C might still hold. The formal version follows.

Theorem 1 (Evidential Vacuity of Accounting Identities)**.** *Let E be an accounting identity and C any claim. Then*

$$E \cap C = C \qquad \text{and} \qquad E \cap \neg C = \neg C.$$

In plain words: the overlap of E with C is C itself, and the overlap of E with "not C" is "not C" itself. Learning E splits Ω into exactly the same two parts that $\{C, \neg C\}$ already give. Nothing is ruled out. Nothing is refined.

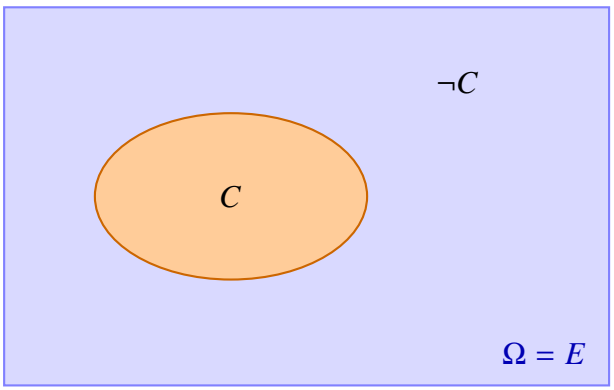
Proof. By Definition 1, $E = \Omega$. Therefore $E \cap C = \Omega \cap C = C$, and $E \cap \neg C = \Omega \cap \neg C = \neg C$. The two splits of Ω coincide. □ □

Figure 2 shows the content of the theorem. The rectangle is Ω. The shaded blob is any contingent claim C. The identity E covers the entire rectangle. Overlapping C with E returns C itself. No part of C is removed, and no part of $\neg C$ is removed. The two possibilities before learning E are the two possibilities after. There is no update to perform.

The statement uses only set equality, the overlap operation, and the definition of an identity as "true in every situation." It needs no probability theory and no likelihood calculus of any kind. Those are introduced separately in Section 5 for readers who want them.

5. Bayesian Restatement

This section restates the theorem in the language of probability, for readers who want that form. Nothing in it is used elsewhere. A reader who takes the main theorem on its set-theoretic terms may skip this section without loss.



$$E \cap C = C \quad \text{and} \quad E \cap \neg C = \neg C$$

Figure 2: The accounting identity E equals the entire state space Ω (the shaded blue rectangle). Any contingent claim C (orange ellipse) and its complement $\neg C$ are unchanged by overlap with E. Learning E rules out no situation, refines no split, and leaves our view of C exactly where it was. This is Theorem 1 in one picture.

We assume that probabilities can be assigned to claims in the usual way.[1] Let $\mathbb{P}$ denote such a probability assignment.

Definition 5 (Evidence)**.** *In plain words: a claim E is* evidence for *a claim C when learning that E holds changes the probability of C. If knowing E leaves the probability of C unchanged, then E carries no information about C.*

In symbols, E is evidence for C under $\mathbb{P}$ when

$$\mathbb{P}(C \mid E) \neq \mathbb{P}(C).$$

The bar "$\mid$" reads "given." $\mathbb{P}(C \mid E)$ is the probability of C given that we already know E. $\mathbb{P}(C)$ is the probability of C before we know E. The condition says these two differ.

This is equivalent to requiring that the ratio

$$\Lambda(E; C, \neg C) := \frac{\mathbb{P}(E \mid C)}{\mathbb{P}(E \mid \neg C)}$$

differ from one. In plain words: the ratio Λ compares the probability of E in a situation where C is true against the probability of E in a situation where C is false. When the two are equal, the ratio is one, and E is equally likely under either outcome: E carries no information about C. When they differ, the ratio is not one, and E shifts the probability of C one way or the other. The equivalence to $\mathbb{P}(C \mid E) \neq \mathbb{P}(C)$ is standard; see Jaynes (2003) [12].

Corollary 1 (Bayesian form of the Theorem)**.** *If E is an accounting identity and $0 < \mathbb{P}(C) < 1$, then*

$$\mathbb{P}(C \mid E) = \mathbb{P}(C), \qquad \Lambda(E; C, \neg C) = 1.$$

In plain words: the probability of C after learning E equals the probability of C before learning E, and the likelihood ratio comparing the two sides equals one. No accounting identity is evidence for any claim.

Proof. From the Theorem, $\mathbb{P}(E \cap C) = \mathbb{P}(C)$, so $\mathbb{P}(E \mid C) = 1$. Symmetrically, $\mathbb{P}(E \mid \neg C) = 1$. The ratio equals one, and Bayes' theorem gives $\mathbb{P}(C \mid E) = \mathbb{P}(C)$. □ □

[1] Formally, we take a collection of claims closed under complement and finite union, and a probability assignment $\mathbb{P}$ on that collection. No measure-theoretic subtlety enters.

6. Two Direct Consequences

Two further facts follow from the theorem. The first says an identity cannot pick a direction between two claims. The second says piling up identities does not help.

Lemma 1 (Direction blindness)**.** *For any two claims C_1 and C_2, an accounting identity E leaves each unchanged: $E \cap C_i = C_i$ for $i = 1, 2$. No accounting identity can favor one claim over another, in particular a causal claim over its reverse.*

Proof. $E = \Omega$ gives $E \cap C_i = C_i$ for each i. □ □

In plain words: a bookkeeping constraint cannot tell debit from credit as cause. Conservation of mass in a two-reservoir system holds under "reservoir A pumps into reservoir B," under "reservoir B pumps into reservoir A," and under any mixture of the two. Any attempt to derive causal direction from the identity alone commits the error of affirming a statement that is already true in every situation.

Lemma 2 (Aggregation of identities)**.** *Let $E_1, E_2, \ldots, E_n$ each be an accounting identity. Then $E_1 \cap E_2 \cap \cdots \cap E_n = \Omega$. In plain words: the overlap of any number of accounting identities is still the whole state space, because each identity is already the whole state space and overlapping the whole with itself changes nothing. By the Theorem, the stack rules out no situation and is no evidence for any causal claim.*

Proof. $E_i = \Omega$ for each i by Definition 1. Overlapping anything with Ω returns that thing (Section 2, fact 1). Therefore $E_1 \cap \cdots \cap E_n = \Omega$. Apply the Theorem. □ □

In plain words: the "many lines of evidence" defense fails when each line is an identity. Stacking n accounting identities yields the same $E = \Omega$ as any single one. Zero multiplied by n is zero. If any of the n items are genuine observations (strictly smaller than Ω), then those are the only pieces that carry any empirical content; the identity terms rule out no situation, carry no information, and can be deleted. The right form of such an argument is therefore "this one observation implies such-and-such," not "n lines jointly imply such-and-such."

The point is not new. It is a consequence of the elementary structure of Bayesian confirmation, formalized by Carnap (1950) [3]: claims with prior probability 1 have degree of confirmation 1 against every evidence set [12]. The value of restating it is its recurrent violation in applied carbon-cycle writing, where variants of (3) are invoked to attribute the sign or size of F_{nat} from measurements of F_{anth} and $\mathrm{d}M_a/\mathrm{d}t$ alone.

7. The Theorem Applied Across Fields

The logical point named in this theorem has been recognized, sometimes as slogan and sometimes as technical warning, in every field that uses conservation or definitional relations. A partial catalogue follows. Each instance obeys Theorem 1.

Economics

Standard accounting identities include the national-income identity $Y \equiv C + I + G + NX$, the closed-economy saving-investment identity $S \equiv I$, the fundamental accounting equation Assets ≡ Liabilities+Equity, and the balance-of-payments identity. Felipe & McCombie (2013) [9] show at book length that regressions supposedly estimating aggregate production functions actually recover the underlying accounting identity rather than any real technological relationship. Christodoulou (2017) [4] formalizes the *accounting identity trap*: when stocks and flows are

mixed, the data matrix lacks full rank, so regressions mixing them cannot be solved without outside constraints. The related *accounting semi-identity* problem shows that ordinary least-squares regression on an incomplete identity mixes arithmetic necessity with any behavioral content, and the closer the data lie to the identity, the less the regression can speak to cause. Simon (1979) [17] noted in his Nobel address that an estimated Cobb-Douglas production function can be a byproduct of the linear identity Value of output ≡ Wage bill + Capital costs. Dorman (2007) [7] stated the point most cleanly: appealing to the identity as evidence for a model is wrong because "there is nothing to be explained." All are instances of Theorem 1.

Medicine and Physiology

The energy-balance equation

$$\Delta E \;=\; E_{\text{in}} - E_{\text{out}}$$

is the first law of thermodynamics applied to the body. It is an accounting identity. The claim "obesity is caused by energy imbalance" is the standard misreading, criticized as a tautology by Taubes (2007) [18], by Fung (2016) [11], and formally by Torres-Carvallo et al. (2022) [19]. Lemma 1 applies directly: the algebraic rewriting $E_{\text{in}} = \Delta E + E_{\text{out}}$ does not pick intake over storage or expenditure as cause. Clinical mass, fluid, and nitrogen balances, and pharmacokinetic ADME mass-balance studies, are accounting inventories; they test completeness of measurement but do not by themselves establish which pathway caused clearance.

Demography, Fluid Dynamics, Bookkeeping

The demographic identity ΔPop = Births − Deaths + Net migration, the continuity and mass-conservation equations of fluid dynamics and atmospheric science, and double-entry bookkeeping itself are all accounting identities in the sense of Definition 1. Each is useful as a consistency constraint. Each is useless, taken alone, as evidence for a directed causal claim. The carbon-cycle mass-balance identity is the next instance.

8. The Carbon Mass-Balance Identity

Corollary 2. *Equation* (3) *is derivable from the conservation postulate* (1) *and the fossil/non-fossil split* (2) *by algebraic rearrangement alone. It holds in every physically allowed situation. It is therefore an accounting identity in the sense of Definition 1.*

Proof. Substituting (2) into (1) and rearranging yields (3). Every situation consistent with (1) and (2) satisfies (3). Hence the collection of situations in which (3) holds is all of Ω. □ □

By Theorem 1, (3) carries zero evidential weight for or against any claim about the source, sink, direction, or size of the non-fossil-fuel flux F_{nat}. Any inference of the form "(3) holds, therefore $F_{\text{nat}} < 0$ on decadal timescales," or "(3) holds, therefore anthropogenic emissions cause the observed $\mathrm{d}M_a/\mathrm{d}t$," is a logical error. The identity carries no causal implication by construction. It constrains only the difference $F_{\text{out}} - F_{\text{nat}}$, both purely definitional quantities, given the measured quantity M_a and the estimated quantity F_{anth}. ■

9. What the Theorem Does Not Forbid: Observations

The Theorem applies to a claim $E = \Omega$ taken in isolation. It does not forbid confirmation of causal claims by *observations* in the sense of Definition 2. An observation fixes the measured quantities to their observed values, is a proper part of Ω, and has a likelihood ratio that need not equal one. Observations may therefore serve as evidence. Identities cannot. What the Theorem forbids is the

inference from any $E = \Omega$ to a causal conclusion. The individual observations that feed into an identity do not inherit its emptiness; only the identity does.

This distinction is critical in application. An argument of the form "we measured M_a, we estimated F_{anth}, we invoked (3) to fix $F_{\text{out}} - F_{\text{nat}}$, and this counts as evidence for anthropogenic attribution" reduces to invocation of the identity and falls under the Theorem: fixing a definitional quantity from measured quantities by algebraic rearrangement adds no claim beyond $E = \Omega$. An argument of the form "we independently measured F_{nat} by isotopic partitioning and obtained the value $\mathbf{m}^*$" promotes F_{nat} from a definitional quantity to a measured quantity and yields an observation that is a proper part of Ω, which must be judged on its own likelihood terms.

10. Application: The Total Failure of Engelbeen, Hannon, and Burton (2024)

The EHB (2024) report [8] purports to demonstrate "conclusively" that the modern rise in atmospheric CO_2 is anthropogenic. Its central chapter builds on the carbon mass balance and returns to the identity in each of the successive sub-arguments audited below. The central chain reduces to the mass-balance identity (3) and its algebraic rearrangements, with one additional statistical move (Violation 3) that is not an identity but is also evidentially empty for a different, equally fatal reason. Under Theorem 1, the identity-based arguments carry exactly zero evidential content for the stated causal conclusion. We enumerate the five violations.

The report's remaining sections advance isotopic, oxygen, and ocean-chemistry arguments. These are observations in the sense of Definition 2, not accounting identities, and are not audited here. They are audited in detail in a separate companion paper, Cohler & Soon (2026a) [5], which shows that they too fail to support EHB's attribution and in fact falsify it. Their status is immaterial to the verdict below in any case. EHB's proof claim rests on the mass balance. The isotopic, oxygen, and ocean-chemistry sections are what EHB themselves label "additional evidence for the human contribution." The mass balance is the proof they offer, in their own words: it "definitely proves" human causation. Theorem 1 nullifies the mass-balance proof. No amount of additional evidence, however strong on its own terms, can convert a null proof into a proof. The paper fails at what it set out to do, regardless of what those sections contain.

10.1 Violation 1: The Mass-Balance Identity Presented as Causal Proof

The opening of EHB's central chapter (p. 7) states: "The carbon mass balance is what definitely proves that humans are the main cause of the increase of CO_2 in the atmosphere: as long as the increase in the atmosphere is less than what humans emit per year, nature as a whole is a net sink and can't be the cause of the overall increase because that would violate the mass balance." The Mass Balance Conclusions (p. 11) repeat the claim in general form: "Any theory that leads to a substantial increase of CO_2 in the atmosphere due to natural factors violates the mass balance and thus cannot be correct." The argument chain runs: $F_{\text{anth}} > \mathrm{d}M_a/\mathrm{d}t$ (the rate of human emission exceeds the rate at which the atmosphere is gaining CO_2), therefore $F_{\text{out}} > F_{\text{nat}}$ (non-fossil absorption exceeds non-fossil emission), therefore nature is a net sink, therefore anthropogenic emissions cause the observed increase. Every step in this chain is a rearrangement of (3). By Definition 1, (3) holds in every situation, including situations in which the non-fossil-fuel flux is the dominant driver and situations in which it is not. By Theorem 1, no such rearrangement can serve as evidence for either causal hypothesis. EHB's "that would violate the mass balance" is not a physical prohibition against non-fossil-fuel causation; it is an assertion that a definitional identity holds, which it does in every situation. The identity forbids nothing about causation. It is not "what definitely proves" anything.

10.2 Violation 2: The Airborne-Fraction Substitution

EHB report that "the trend of the CO_2 increase in the atmosphere is around 50% of human emissions" and immediately conclude: "That proves that nature is a sink for the other 50%, not the cause of the increase." The reported ratio is

$$\mathrm{ABF_C} \;:=\; \frac{\mathrm{d}M_a/\mathrm{d}t}{F_{\mathrm{anth}}},$$

a flow ratio derived directly from (3). In plain words: the numerator is the rate at which the atmospheric CO_2 mass is currently rising, and the denominator is the rate at which humans are currently emitting CO_2. The subscript C on $\mathrm{ABF_C}$ marks this as what Koutsoyiannis (2024b) [14] calls the *Computational* airborne fraction (he uses C for computational and L for logical, which we follow throughout). This is an accounting quantity: given F_{anth} and $\mathrm{d}M_a/\mathrm{d}t$, $\mathrm{ABF_C}$ is fixed by arithmetic. It carries no information about the fraction of anthropogenic emissions retained in the atmosphere and is not the retention fraction $\mathrm{ABF_L}$, which the IPCC AR6 Glossary of Matthews et al. (2021) [15] defines as "the fraction of total CO_2 emissions (from fossil fuel and land use change) remaining in the atmosphere." In plain words: the subscript L on $\mathrm{ABF_L}$ marks this as the *Logical* airborne fraction in Koutsoyiannis's convention, meaning cumulative anthropogenic CO_2 still in the atmosphere divided by total anthropogenic CO_2 ever emitted. Under proper mass-balance reservoir routing, per Koutsoyiannis (2024a) [13], $\mathrm{ABF_L} \approx 6\%$ with residence time $\tau_{\mathrm{res}} \approx 3.8$ yr (the arithmetic average of the Mauna Loa and Barrow theoretical-mean values of 3.68 yr and 3.91 yr reported in Table 4 of that paper). The symbol τ_{res} stands for residence time: the average number of years a CO_2 molecule spends in the atmosphere before leaving through a sink. The residence time follows directly from the standard hydrological definition of outflow time scale for a reservoir, namely reservoir mass divided by gross outflow rate:

$$\tau_{\mathrm{res}} \;=\; \frac{M_a}{F_{\mathrm{out}}}.$$

Dividing the global atmospheric CO_2 inventory implied by the Mauna Loa mole fraction averaged over 2010–2019 by the gross outflow flux F_{out} tabulated in IPCC AR6 Figure 5.12 [2] yields $\tau_{\mathrm{res}} \approx 3.83$ yr independently. Two independent routes converge on the same figure. EHB substitute the accounting ratio $\mathrm{ABF_C}$ for the physical retention fraction $\mathrm{ABF_L}$ without disclosure. The substitution presents an arithmetic quantity as a physical constant, and the arithmetic is itself an instance of Theorem 1.

10.3 Violation 3: Cumulative Correlation of Monotonic Series

EHB display a correlation of $R^2 \approx 0.998$ between cumulative anthropogenic emissions and atmospheric CO_2 concentration and treat the figure as evidence for causation. Both series are integrals of positive quantities and are therefore monotonically increasing over the modern interval. Any two series that share a common monotonic trend show high positive correlation as an arithmetic consequence of that shared trend. The specific numerical value is not fixed by monotonicity alone, but values near 1 are the expected result whenever the relative rates of increase are roughly stable.

The observed R^2 is a contingent empirical fact about the two particular time paths. It is not forced by the mass-balance identity (3). It is not an accounting identity. Theorem 1 does not apply to it. The correlation remains a weak and confounded statistic for causal inference. The dominant contribution to the raw R^2 is the common long-term rise. Once that rise is removed (for example by fitting a separate cubic polynomial to each series and correlating the residuals), the residual association is the quantity relevant to any claim of co-movement beyond the shared trend. The undetrended figure supplies no such information.

The inference from the raw correlation to anthropogenic causation is therefore invalid. It rests on the classic confounding of two trending series rather than on an accounting identity. The logical

defect is the same in kind: a statistic that is expected under the mere positivity and rough stability of the two cumulative paths is treated as evidence that one path causes the other.

10.4 Violation 4: "Nature Is a Net Sink" as Independent Argument

EHB present the inference "$F_{\text{nat}} - F_{\text{out}} < 0$, therefore nature is a net sink, therefore anthropogenic emissions cause the rise" as a separate line of evidence. In plain words: the inequality $F_{\text{nat}} - F_{\text{out}} < 0$ says that non-fossil-fuel emission minus non-fossil-fuel absorption is negative, which is the same as saying absorption exceeds emission on the non-fossil side. This is algebraically equivalent to $F_{\text{anth}} > \mathrm{d}M_a/\mathrm{d}t$ (human emission exceeds the rate at which the atmosphere is gaining CO_2), which is (3) rearranged. The "line of evidence" is the identity rewritten in different variables. By Lemma 1, the identity is direction-blind: a situation in which the non-fossil-fuel flux drives M_a upward with anthropogenic emissions passively absorbed by a large sink satisfies (3) exactly. The identity cannot select between the two.

10.5 Violation 5: The "Multiple Lines of Evidence" Defense

The EHB abstract asserts that "the CO_2 Coalition presents multiple lines of scientific evidence demonstrating conclusively that the modern increase in CO_2 is mainly due to anthropogenic emissions." The invocation of "multiple lines" is the standard rhetorical maneuver by which authors deflect the accounting-identity objection: even if one line is challenged, others are said to remain. Under audit, each line in that central chain is either a direct invocation of (3), an algebraic rearrangement of (3), or a numerical restatement of one term in (3) in different units. Each such line equals Ω. By Lemma 2, the overlap of any finite collection of such identities is again Ω, and by the Theorem that stack rules out no situation either. There is no claim $E \neq \Omega$ in the central argument. The apparent multiplicity of evidence is a rhetorical byproduct of restating the same accounting identity in five or six different notations. Zero evidential content stacked across five or six lines remains zero. The "multiple lines" framing is not a reply to the objection raised by Theorem 1; it is a further instance of the error the Theorem forbids.

10.6 Consequence

Every load-bearing step in EHB's mass-balance chain is an accounting identity: a statement true in every possible situation, one that rules none out. By the Theorem, no such step is evidence for human causation, or against it. The chain EHB present as definitive proof is a string of definitional identities dressed up as a proof. It proves nothing.

The report's conclusion, that human emissions have been *proved* to be the main cause of the atmospheric CO_2 rise, is therefore false as a matter of logic. It is not weak evidence. It is not disputed evidence. It is not evidence pending further work. It is no evidence.

The remaining chapters of EHB (isotopic, oxygen, and ocean-chemistry) are real observations, not identities. They could in principle serve as evidence either way. But EHB do not offer them as an alternative proof, and no finite stack of real-world observations can ever be a mathematical proof. More importantly, our companion paper [5] examines those same observations on their own terms and shows that they strongly argue *against* human causation, not for it. The observational evidence EHB cite in support of human dominance, read correctly, points the other way.

EHB's proof fails by the Theorem. The observational evidence they rely on, properly analyzed, falsifies the very claim they set out to prove.

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Conflict of Interest

The authors declare no conflict of interest.

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